Figure 1: The National LIbrary of Medicine PubMed Central Article Dataset, ie. the corpus.

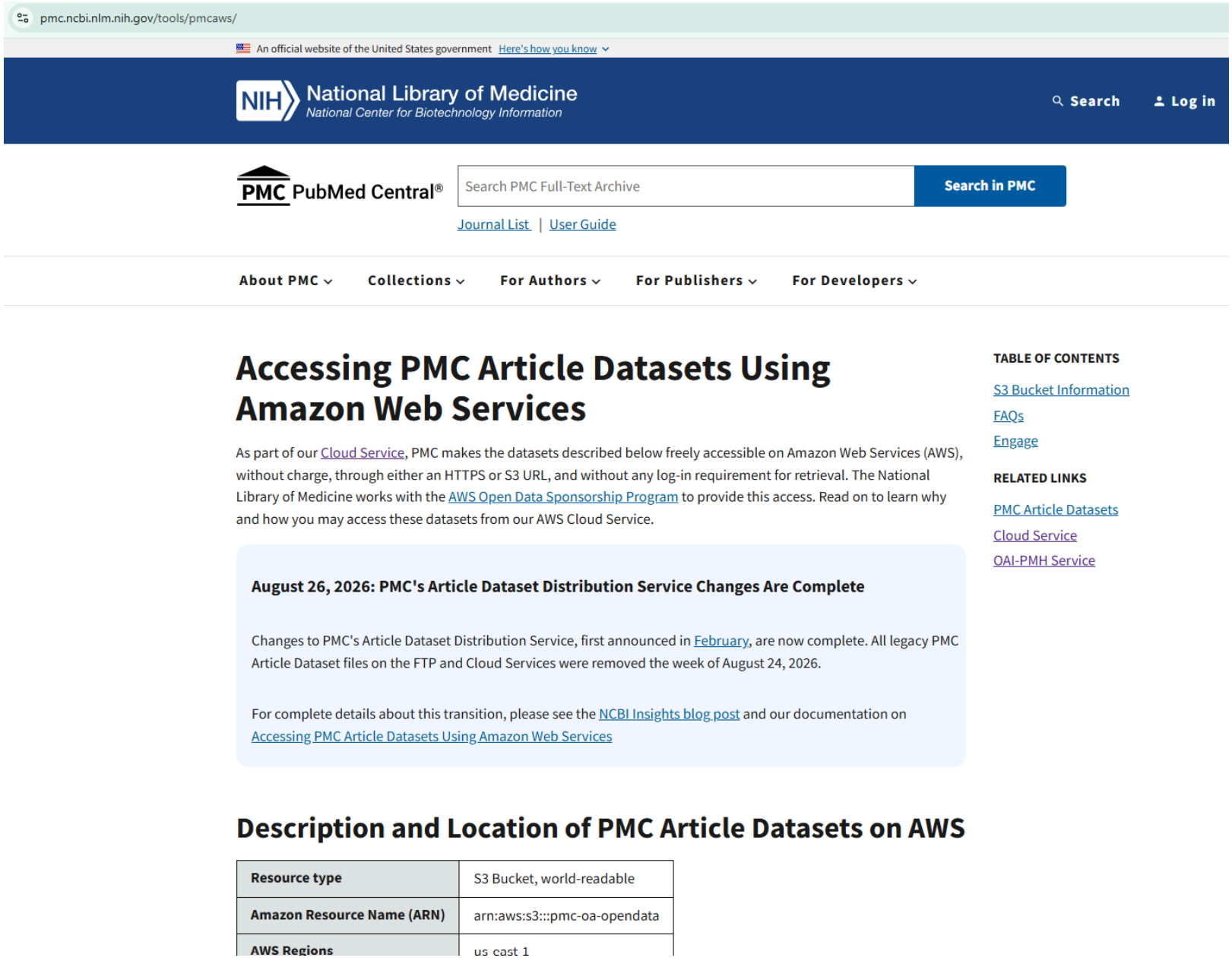


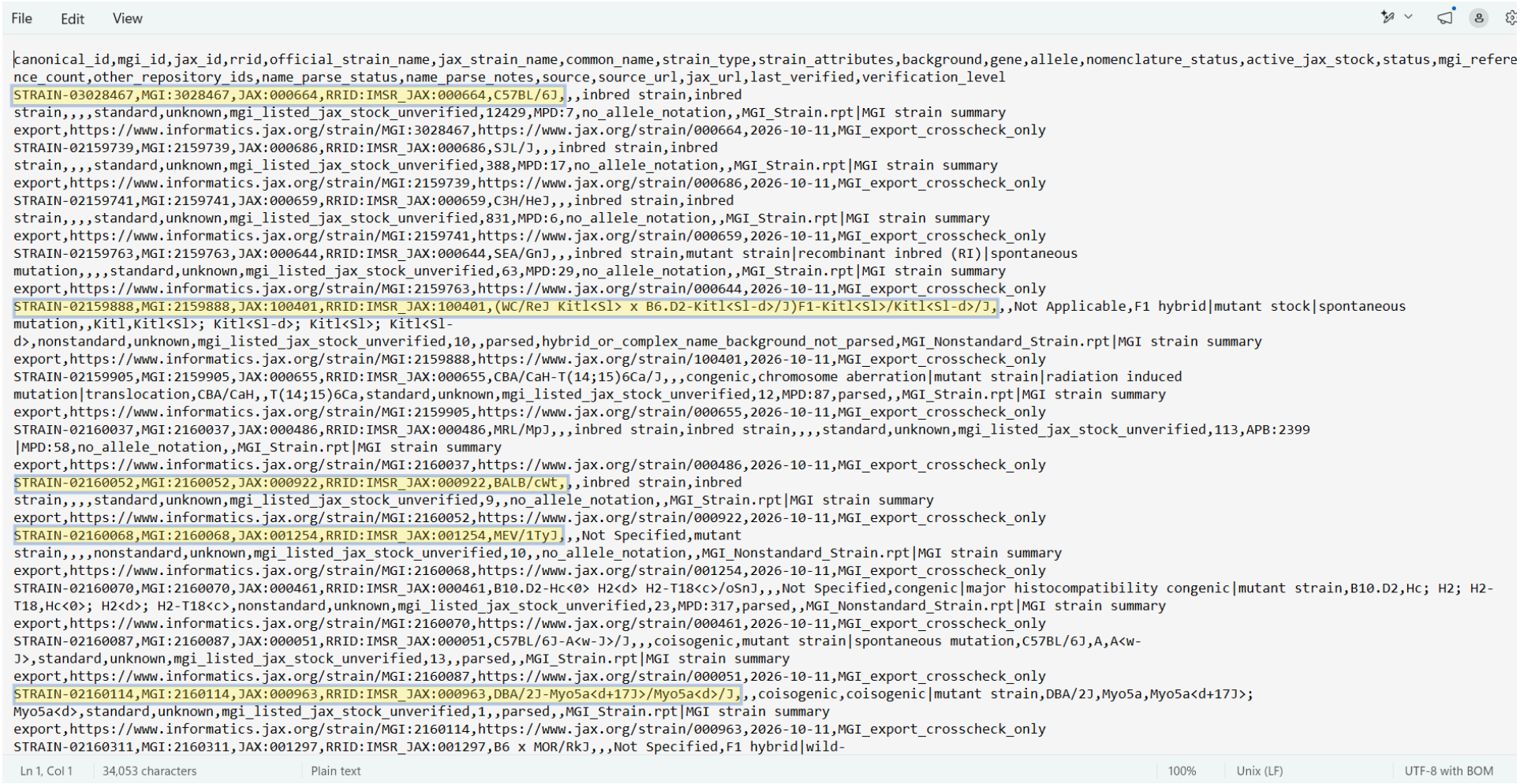
Figure 2: Table of 7 possible designations used to identify product

Name, label, or alias	Example
strain_key	STRAIN-000001
mgi_id	MGI:3028467
official_mgi_name	C57BL/6J
strain_type	inbred strain
jax_stock_id	JAX:000664
rrid	RRID:IMSR_JAX:000664
jax_name	C57BL/6J
common_name	C57BL/6

The AI solution documented here extracts citations of actual use of product explicitly mentioned or implicitly linked in scientific publications. The challenge is identification accuracy where a great deal of non-standard, variant nomenclature exists. The 1st Pubmed Pipeline solution that I developed in 2023 for the Jackson Laboratory reached the "end-of-life" state because Amazon Web Services discontinued its Kendra service in August 2026. I have taken this opportunity to rebuild and update the solution using the current, state-of-the-AI-art tools in order to accomplish the same functionality. I am no longer bound by the same organizational Intellectual Property constraints since I left that employment. In addition, the Jackson Laboratory's published lexicon of nomenclature, names, and aliases as well as the PubMed Central corpus of scientific publications both exist in the public domain.

Please find my agentic AI sample in the following public GitHub repo: https://github.com/cloudcounselor/pubmed-pipeline

Figure 3: Source data published by The Jackson Laboratory to catalog the standard nomenclature for products in the catalog.



File Edit View

```
canonical_id,mgi_id,jax_id,rrid,official_strain_name,jax_strain_name,common_name,strain_type,strain_attributes,background,gene,allele,nomenclature_status,active_jax_stock,status,mgi_refere
nce_count,other_repository_ids,name_parse_status,name_parse_notes,source,source_url,jax_url,last_verified,verification_level
STRAIN-03028467,MGI:3028467,JAX:000664,RRID:IMSR_JAX:000664,C57BL/6J,,,inbred strain,inbred
strain,,,,standard,unknown,mgi_listed_jax_stock_unverified,12429,MPD:7,no_allele_notation,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:3028467,https://www.jax.org/strain/000664,2026-10-11,MGI_export_crosscheck_only
STRAIN-02159739,MGI:2159739,JAX:000686,RRID:IMSR_JAX:000686,SJL/J,,,inbred strain,inbred
strain,,,,standard,unknown,mgi_listed_jax_stock_unverified,388,MPD:17,no_allele_notation,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2159739,https://www.jax.org/strain/000686,2026-10-11,MGI_export_crosscheck_only
STRAIN-02159741,MGI:2159741,JAX:000659,RRID:IMSR_JAX:000659,C3H/HeJ,,,inbred strain,inbred
strain,,,,standard,unknown,mgi_listed_jax_stock_unverified,831,MPD:6,no_allele_notation,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2159741,https://www.jax.org/strain/000659,2026-10-11,MGI_export_crosscheck_only
STRAIN-02159763,MGI:2159763,JAX:000644,RRID:IMSR_JAX:000644,SEA/GnJ,,,inbred strain,mutant strain|recombinant inbred (RI)|spontaneous
mutation,,,,standard,unknown,mgi_listed_jax_stock_unverified,63,MPD:29,no_allele_notation,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2159763,https://www.jax.org/strain/000644,2026-10-11,MGI_export_crosscheck_only
STRAIN-02159888,MGI:2159888,JAX:100401,RRID:IMSR_JAX:100401,(WC/ReJ Kitl<Sl> x B6.D2-Kitl<Sl-d>/J)F1-Kitl<Sl>/Kitl<Sl-d>/J,,,Not Applicable,F1 hybrid|mutant stock|spontaneous
mutation,,Kitl,Kitl<Sl>; Kitl<Sl-d>; Kitl<Sl>; Kitl<Sl-
d>,nonstandard,unknown,mgi_listed_jax_stock_unverified,10,,parsed,hybrid_or_complex_name_background_not_parsed,MGI_Nonstandard_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2159888,https://www.jax.org/strain/100401,2026-10-11,MGI_export_crosscheck_only
STRAIN-02159905,MGI:2159905,JAX:000655,RRID:IMSR_JAX:000655,CBA/CaH-T(14;15)6Ca/J,,,congenic,chromosome aberration|mutant strain|radiation induced
mutation|translocation,CBA/CaH,,T(14;15)6Ca,standard,unknown,mgi_listed_jax_stock_unverified,12,MPD:87,parsed,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2159905,https://www.jax.org/strain/000655,2026-10-11,MGI_export_crosscheck_only
STRAIN-02160037,MGI:2160037,JAX:000486,RRID:IMSR_JAX:000486,MRL/MpJ,,,inbred strain,inbred strain,,,,standard,unknown,mgi_listed_jax_stock_unverified,113,APB:2399
|MPD:58,no_allele_notation,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2160037,https://www.jax.org/strain/000486,2026-10-11,MGI_export_crosscheck_only
STRAIN-02160052,MGI:2160052,JAX:000922,RRID:IMSR_JAX:000922,BALB/cWt,,,inbred strain,inbred
strain,,,,standard,unknown,mgi_listed_jax_stock_unverified,9,,no_allele_notation,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2160052,https://www.jax.org/strain/000922,2026-10-11,MGI_export_crosscheck_only
STRAIN-02160068,MGI:2160068,JAX:001254,RRID:IMSR_JAX:001254,MEV/1TyJ,,,Not Specified,mutant
strain,,,,nonstandard,unknown,mgi_listed_jax_stock_unverified,10,,no_allele_notation,,MGI_Nonstandard_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2160068,https://www.jax.org/strain/001254,2026-10-11,MGI_export_crosscheck_only
STRAIN-02160070,MGI:2160070,JAX:000461,RRID:IMSR_JAX:000461,B10.D2-Hc<0> H2<d> H2-T18<c>/oSnJ,,,Not Specified,congenic|major histocompatibility congenic|mutant strain,B10.D2,Hc; H2; H2-
T18,Hc<0>; H2<d>; H2-T18<c>,nonstandard,unknown,mgi_listed_jax_stock_unverified,23,MPD:317,parsed,,MGI_Nonstandard_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2160070,https://www.jax.org/strain/000461,2026-10-11,MGI_export_crosscheck_only
STRAIN-02160087,MGI:2160087,JAX:000051,RRID:IMSR_JAX:000051,C57BL/6J-A<w-J>/J,,,coisogenic,mutant strain|spontaneous mutation,C57BL/6J,A,A<w-
J>,standard,unknown,mgi_listed_jax_stock_unverified,13,,parsed,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2160087,https://www.jax.org/strain/000051,2026-10-11,MGI_export_crosscheck_only
STRAIN-02160114,MGI:2160114,JAX:000963,RRID:IMSR_JAX:000963,DBA/2J-Myo5a<d+17J>/Myo5a<d>/J,,,coisogenic,coisogenic|mutant strain,DBA/2J,Myo5a,Myo5a<d+17J>;
Myo5a<d>,standard,unknown,mgi_listed_jax_stock_unverified,1,,parsed,,MGI_Strain.rpt|MGI strain summary
export,https://www.informatics.jax.org/strain/MGI:2160114,https://www.jax.org/strain/000963,2026-10-11,MGI_export_crosscheck_only
STRAIN-02160311,MGI:2160311,JAX:001297,RRID:IMSR_JAX:001297,B6 x MOR/RkJ,,,Not Specified,F1 hybrid|wild-
```

Ln 1, Col 1 34,053 characters Plain text 100% Unix (LF) UTF-8 with BOM

Figure 4: PowerShell interface: creation of Database schema objects in PostgreSQL.

```
PS C:\Users\Mary\Documents\pythonapps\pubmed-pipeline> docker exec -it pubmed_postgres psql -U postgres -d pubmed_analytics
psql (15.19)
Type "help" for help.

pubmed_analytics=# \conninfo
You are connected to database "pubmed_analytics" as user "postgres" via socket in "/var/run/postgresql" at port "5432".
pubmed_analytics=# \dt public.*
                List of relations
 Schema |          Name          | Type  |  Owner
--------+------------------------+-------+----------
 public | human_review_audit_log | table | postgres
 public | jax_products           | table | postgres
 public | publication_mentions   | table | postgres
(3 rows)
```

```
pubmed_analytics=# \d+ public.jax_products
                                                  Table "public.jax_products"
     Column      |           Type           | Collation | Nullable |           Default            | Storage  | Compression | Stats target | Description
-----------------+--------------------------+-----------+----------+------------------------------+----------+-------------+--------------+-------------
 jax_id          | character varying(64)    |           | not null |                              | extended |             |              |
 product_name    | character varying(255)   |           |          |                              | extended |             |              |
 expected_status | character varying(32)    |           |          | 'UNKNOWN'::character varying | extended |             |              |
 created_at      | timestamp with time zone |           |          | CURRENT_TIMESTAMP            | plain    |             |              |
 updated_at      | timestamp with time zone |           | not null | CURRENT_TIMESTAMP            | plain    |             |              |
Indexes:
    "jax_products_pkey" PRIMARY KEY, btree (jax_id)
Referenced by:
    TABLE "publication_mentions" CONSTRAINT "publication_mentions_jax_id_fkey" FOREIGN KEY (jax_id) REFERENCES jax_products(jax_id) ON DELETE CASCADE
Triggers:
    update_jax_products_modtime BEFORE UPDATE ON jax_products FOR EACH ROW EXECUTE FUNCTION update_modified_column()
Access method: heap
```

Figure 5: PowerShell interface showing product data (abridged) loaded into the PostgreSQL “jax_products” table.

```
pubmed_analytics-# FROM public.jax_products
pubmed_analytics-# ORDER BY jax_id
pubmed_analytics-# LIMIT 25;
   jax_id   |                product_name                | expected_status |         created_at
------------+--------------------------------------------+-----------------+-------------------------------
 JAX:000021 | B6.Cg-A<y>/J                               | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000024 | B6.Cg-Bloc1s6<pa>/J                        | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000051 | C57BL/6J-A<w-J>/J                          | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000094 | 129/Sv                                     | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000102 | B6.C3-Ap3b1<pe>/J                          | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000103 | B6.Cg-Hps6<ru>/J                           | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000136 | B6.C-H34<c>/(HW22)ByJ                      | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000142 | B6.C-Ifi44l<H28-c> If1<l>/(HW97)cByJ       | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000160 | B6.D2-Kitl<Sl-d>/J                         | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000172 | MK/ReJ                                     | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000223 | C3H/HeJ-Mgrn1<md>/J                        | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000226 | B6C3Fe a/a-Large1<myd>/J                   | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000256 | B6.C-H2-K<bm1>/ByBir-Gusb<mps>/J           | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000348 | C57BL/6J-Sgk3<fz> Mlph<ln>                 | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000350 | B6By.Cg-Kit<W-v> Mitf<Mi-wh> T/J           | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000368 | B6.C-H2-K<bm1>/By                          | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000409 | B10.129P3-H1<b> Hbb<d> Tyr<c> Ea7<a>/(5M)oSnJ | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000419 | B10.UW-H3<b> Tgm3<we> Pax1<un> a<t>/SnJ    | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000435 | D1.LP-H2<b> H2-T18<b?>/SnJ                 | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000461 | B10.D2-Hc<0> H2<d> H2-T18<c>/oSnJ          | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000486 | MRL/MpJ                                    | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000518 | B6.Cg-Usp14<ax-J>/J                        | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000568 | B6.Cg-Pmp22<Tr-J> Krt25<Re>/+ +/J          | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000609 | RBB/DnJ                                    | UNKNOWN         | 2026-10-21 03:02:47.298982+00
 JAX:000611 | RBF/Dn                                     | UNKNOWN         | 2026-10-21 03:02:47.298982+00
(25 rows)
```

Figure 6: PowerShell interface showing attributes and metadata for the PostgreSQL “publications_mentions” table. This table serves as the destination for all search-results data captured deterministically by the “s3_chunk_scanner_gpt.py” Python program.

```
pubmed_analytics=# \d+ public.publication_mentions
                                                  Table "public.publication_mentions"
       Column        |           Type           | Collation | Nullable |               Default               | Storage  | Compression | Stats target | Description
---------------------+--------------------------+-----------+----------+-------------------------------------+----------+-------------+--------------+-------------
 mention_id          | uuid                     |           | not null | uuid_generate_v4()                  | plain    |             |              |
 jax_id              | character varying(64)    |           | not null |                                     | extended |             |              |
 s3_uri              | text                     |           | not null |                                     | extended |             |              |
 matched_text        | character varying(128)   |           | not null |                                     | extended |             |              |
 context_window      | text                     |           | not null |                                     | extended |             |              |
 is_valid_jax_mention | boolean                  |           |          | false                               | plain    |             |              |
 product_status      | character varying(32)    |           |          |                                     | extended |             |              |
 confidence_score    | numeric(4,3)             |           |          |                                     | main     |             |              |
 ai_justification    | text                     |           |          |                                     | extended |             |              |
 pipeline_status     | pipeline_status_enum     |           |          | 'PENDING_LLM'::pipeline_status_enum | plain    |             |              |
 error_log           | text                     |           |          |                                     | extended |             |              |
 extracted_at        | timestamp with time zone |           |          |                                     | plain    |             |              |
 created_at          | timestamp with time zone |           |          | CURRENT_TIMESTAMP                   | plain    |             |              |
 updated_at          | timestamp with time zone |           |          | CURRENT_TIMESTAMP                   | plain    |             |              |
 mention_fingerprint | text                     |           | not null |                                     | extended |             |              |
 pmcid               | character varying(16)    |           |          |                                     | extended |             |              |
 article_version     | character varying(16)    |           |          |                                     | extended |             |              |
 publication_title   | text                     |           | not null | ''::text                            | extended |             |              |
 author              | text                     |           | not null | ''::text                            | extended |             |              |
 publication_name    | text                     |           | not null | ''::text                            | extended |             |              |
 publication_date    | text                     |           | not null | ''::text                            | extended |             |              |
 matched_alias       | text                     |           | not null | ''::text                            | extended |             |              |
Indexes:
    "publication_mentions_pkey" PRIMARY KEY, btree (mention_id)
    "idx_pub_mentions_dedup" UNIQUE, btree (jax_id, s3_uri, md5(context_window))
    "idx_pub_mentions_human_review_queue" btree (created_at, mention_id) WHERE pipeline_status = 'HUMAN_REVIEW'::pipeline_status_enum
    "idx_pub_mentions_jax_id" btree (jax_id)
    "idx_pub_mentions_pipeline_status" btree (pipeline_status)
    "idx_pub_mentions_pmcid" btree (pmcid)
    "idx_pub_mentions_publication_date" btree (publication_date)
    "uq_publication_mentions_fingerprint" UNIQUE, btree (mention_fingerprint)
Foreign-key constraints:
    "publication_mentions_jax_id_fkey" FOREIGN KEY (jax_id) REFERENCES jax_products(jax_id) ON DELETE CASCADE
Referenced by:
    TABLE "human_review_audit_log" CONSTRAINT "fk_human_review_audit_mention" FOREIGN KEY (mention_id) REFERENCES publication_mentions(mention_id) ON DELETE RESTRICT
Triggers:
    update_pub_mentions_modtime BEFORE UPDATE ON publication_mentions FOR EACH ROW EXECUTE FUNCTION update_modified_column()
Access method: heap
```

Figure 7: PowerShell interface showing raw output data from agentic AI evaluation of edge cases. This output data is representative of the ambiguous matches that AI directs to the “human review queue”.

```
PS C:\Users\Mary\Documents\pythonapps\pubmed-pipeline> python langgraph_orchestrator.py
OpenAI API key successfully loaded.
Running LangGraph OpenAI evaluation smoke test...
{'jax_id': '000664', 's3_uri': 's3://pmc-oa-opendata/example.xml', 'matched_text': 'C57BL/6J', 'matched_alias': 'C57BL/6J',
 'context_window': 'Mice of the C57BL/6J strain were obtained from The Jackson Laboratory.', 'normalized_term': 'C57BL/6J',
 'n_candidates_jax_set': 1, 'n_candidates_all_mgi': 4, 'is_ambiguous_in_jax_set': False, 'ambiguity_metadata_quality': 'CAT
ALOG_COMPLETE_MGI_AVAILABLE', 'normalized_alias': 'c57bl6j', 'candidate_identifiers': ['JAX:000664', 'MGI:3028467'], 'alias
_total_occurrences': 12, 'alias_distinct_articles': 8, 'evaluation': None, 'routing_action': 'LOG_ERROR', 'error': "OpenAI
evaluation failed: OpenAIRateLimitError: Error code: 429 - {'error': {'message': 'You have no credits remaining. Add credit
s to continue using the API at https://platform.openai.com/settings/organization/billing/.', 'type': 'insufficient_quota',
'param': None, 'code': 'credit_balance_exhausted'}}"}
PS C:\Users\Mary\Documents\pythonapps\pubmed-pipeline> python langgraph_orchestrator.py
OpenAI API key successfully loaded.
Running LangGraph OpenAI evaluation smoke test...
{'jax_id': '000664', 's3_uri': 's3://pmc-oa-opendata/example.xml', 'matched_text': 'C57BL/6J', 'matched_alias': 'C57BL/6J',
 'context_window': 'Mice of the C57BL/6J strain were obtained from The Jackson Laboratory.', 'normalized_term': 'C57BL/6J',
 'n_candidates_jax_set': 1, 'n_candidates_all_mgi': 4, 'is_ambiguous_in_jax_set': False, 'ambiguity_metadata_quality': 'CAT
ALOG_COMPLETE_MGI_AVAILABLE', 'normalized_alias': 'c57bl6j', 'candidate_identifiers': ['JAX:000664', 'MGI:3028467'], 'alias
_total_occurrences': 12, 'alias_distinct_articles': 8, 'evaluation': JAXProductEvaluation(is_valid_jax_mention=True, produc
t_status='ACTIVE', confidence_score=0.98, justification='The sentence explicitly states mice of the C57BL/6J strain were ob
tained from The Jackson Laboratory, indicating current use of the JAX product (JAX:000664). JAX candidate set is unambiguou
s for this normalized term.'), 'routing_action': 'UPSERT_TO_POSTGRES', 'error': ''}
```

Figure 8: PowerShell interface showing examples of AI evaluation, based on interpretation of evidence. The key outcome of this process, controlled by the “langgraph_orchestrator.py” Python script, is the assignment of “pipeline status”.

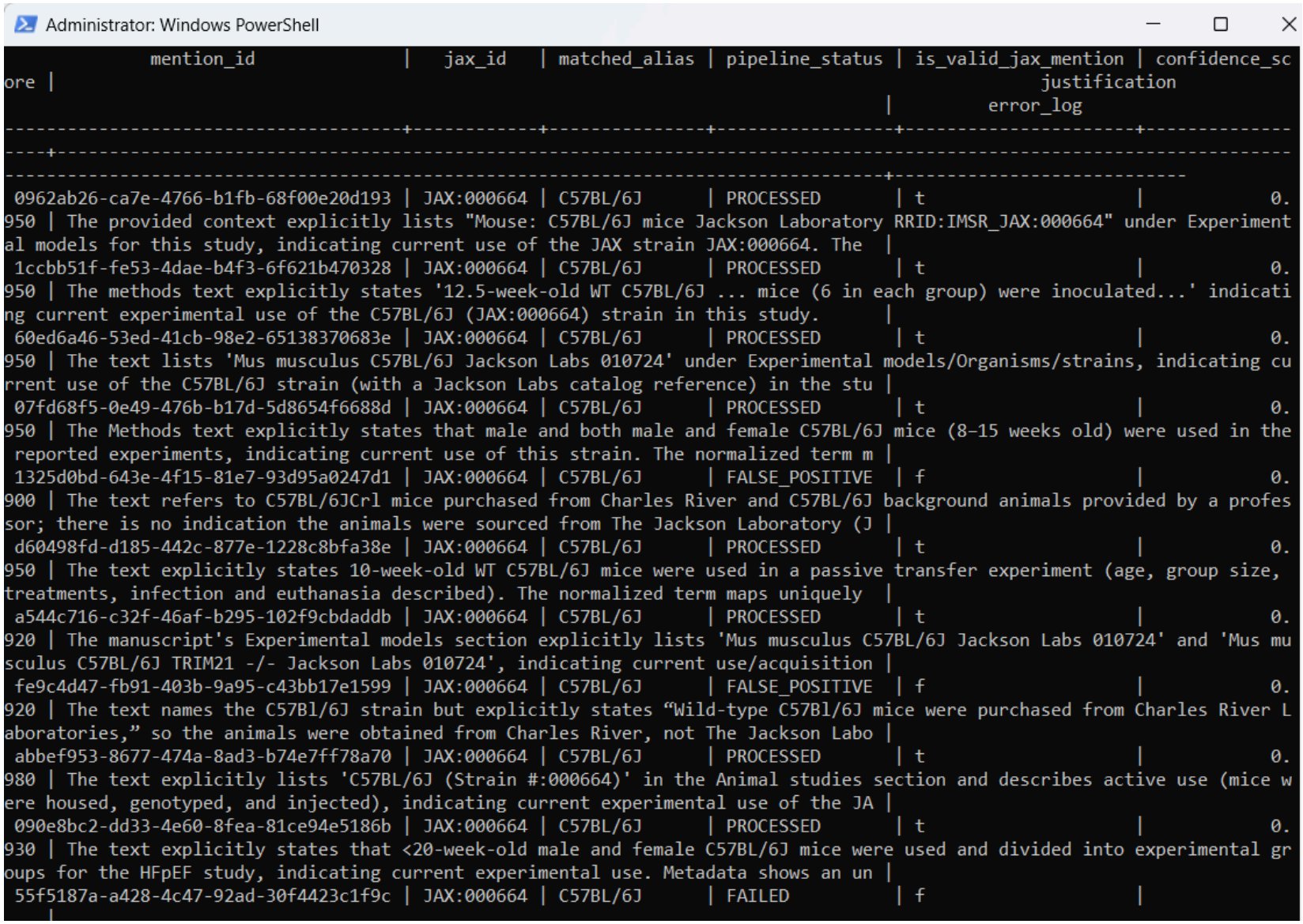


Figure 9: Web browser display showing the StreamLit application interface used for human review, ie. assigning expert judgments to edge cases.

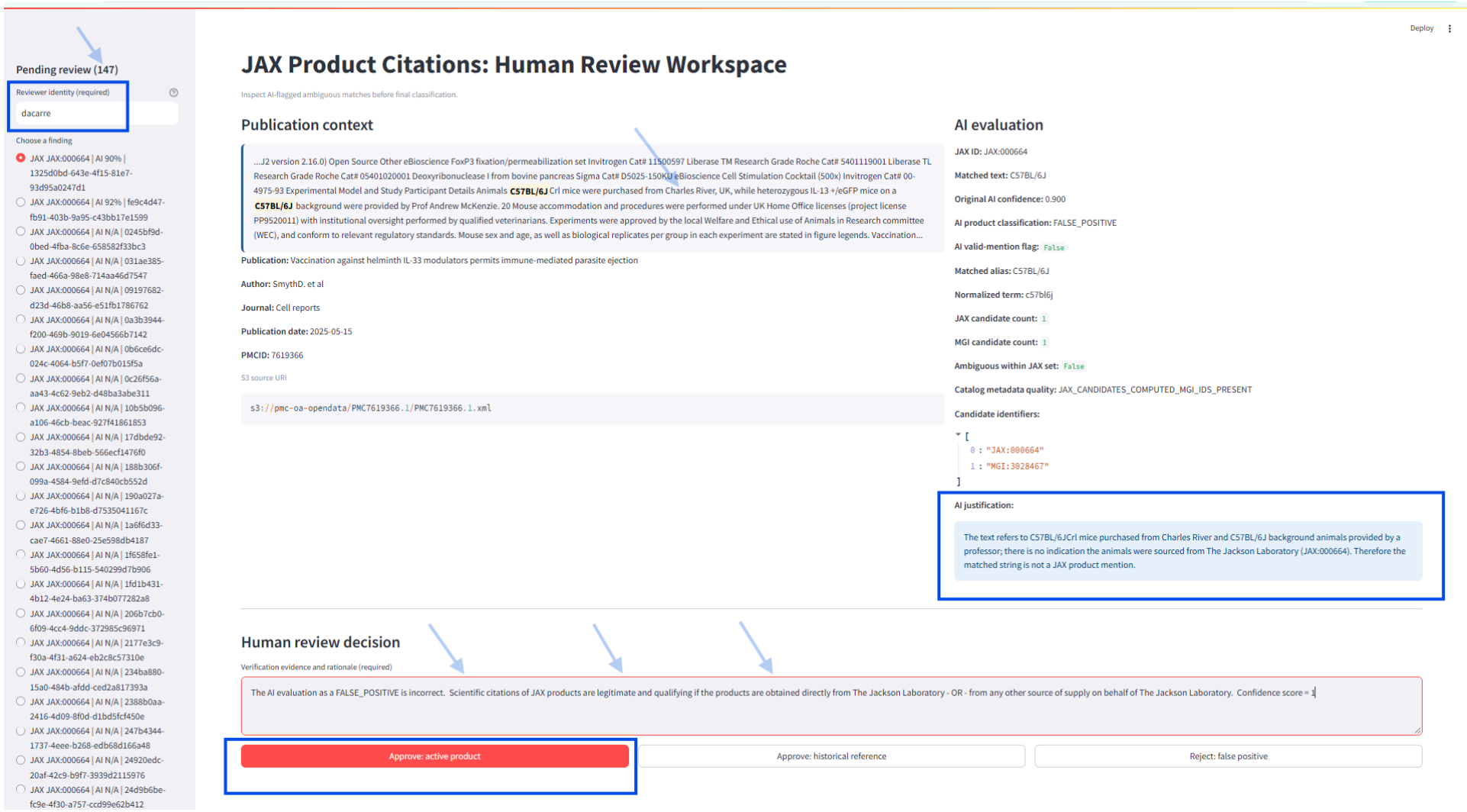


Figure 10: DBeaver interface showing query results from the “pubmed_analytics” database.

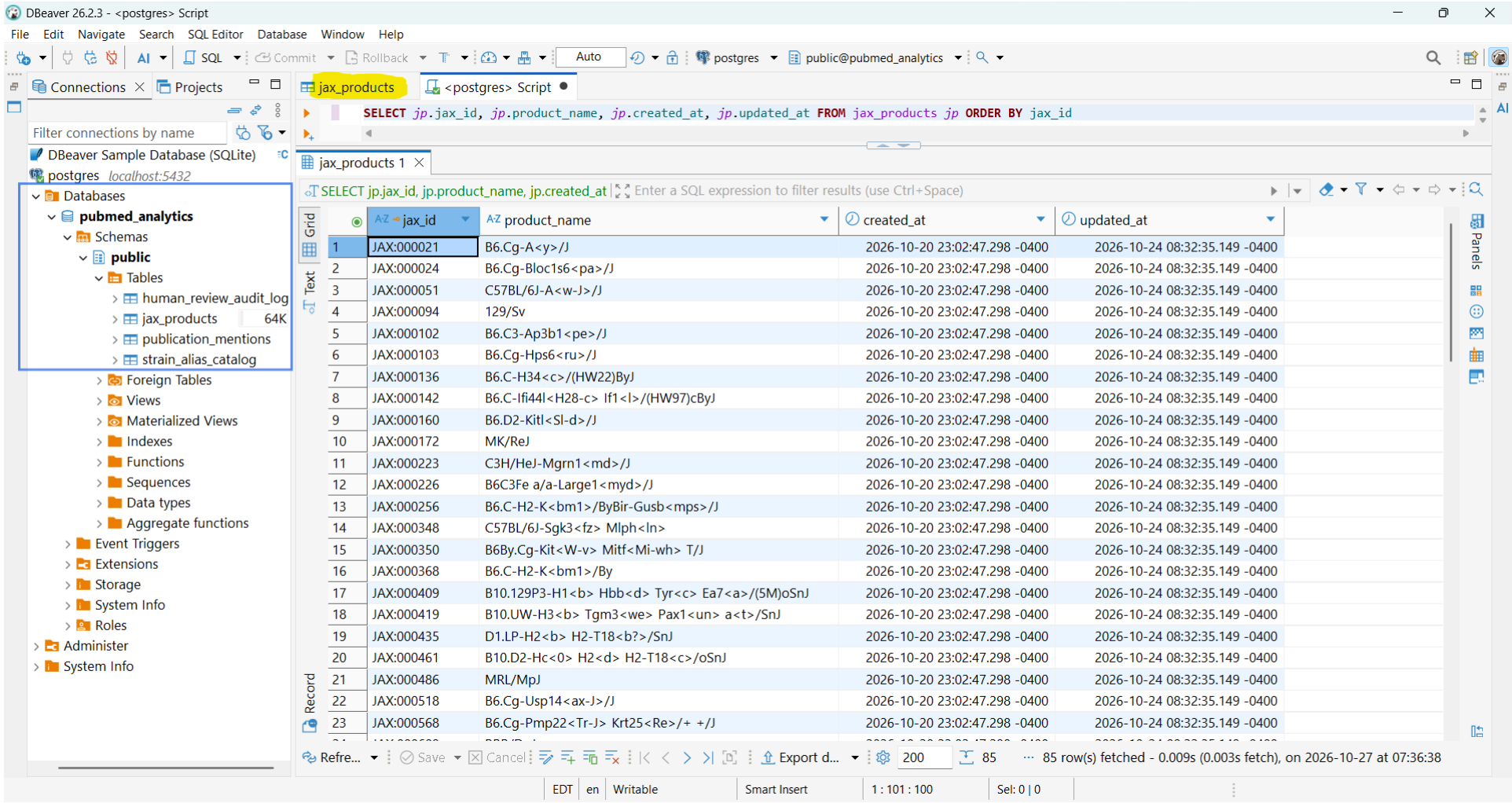


Figure 11: DBeaver interface showing query results from the “strain_alias_catalog” table, which is built deterministically by the “pipeline_coordinator.py” Python script.

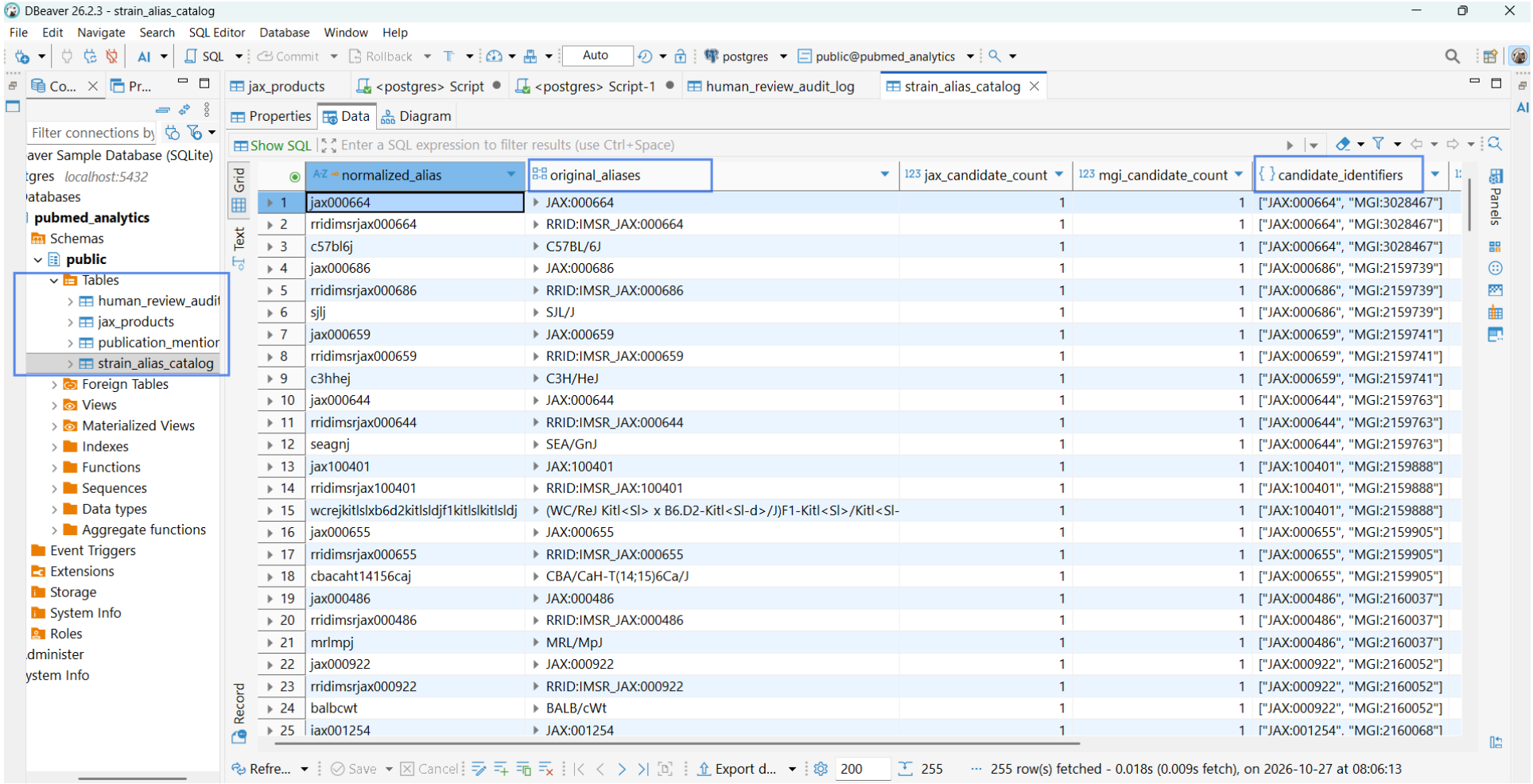


Figure 12: DBeaver interface showing query results from the “publication_mentions” table, which holds all the results of deterministic, non-AI searching that is then routed agentically to AI for probabilistic evaluation.

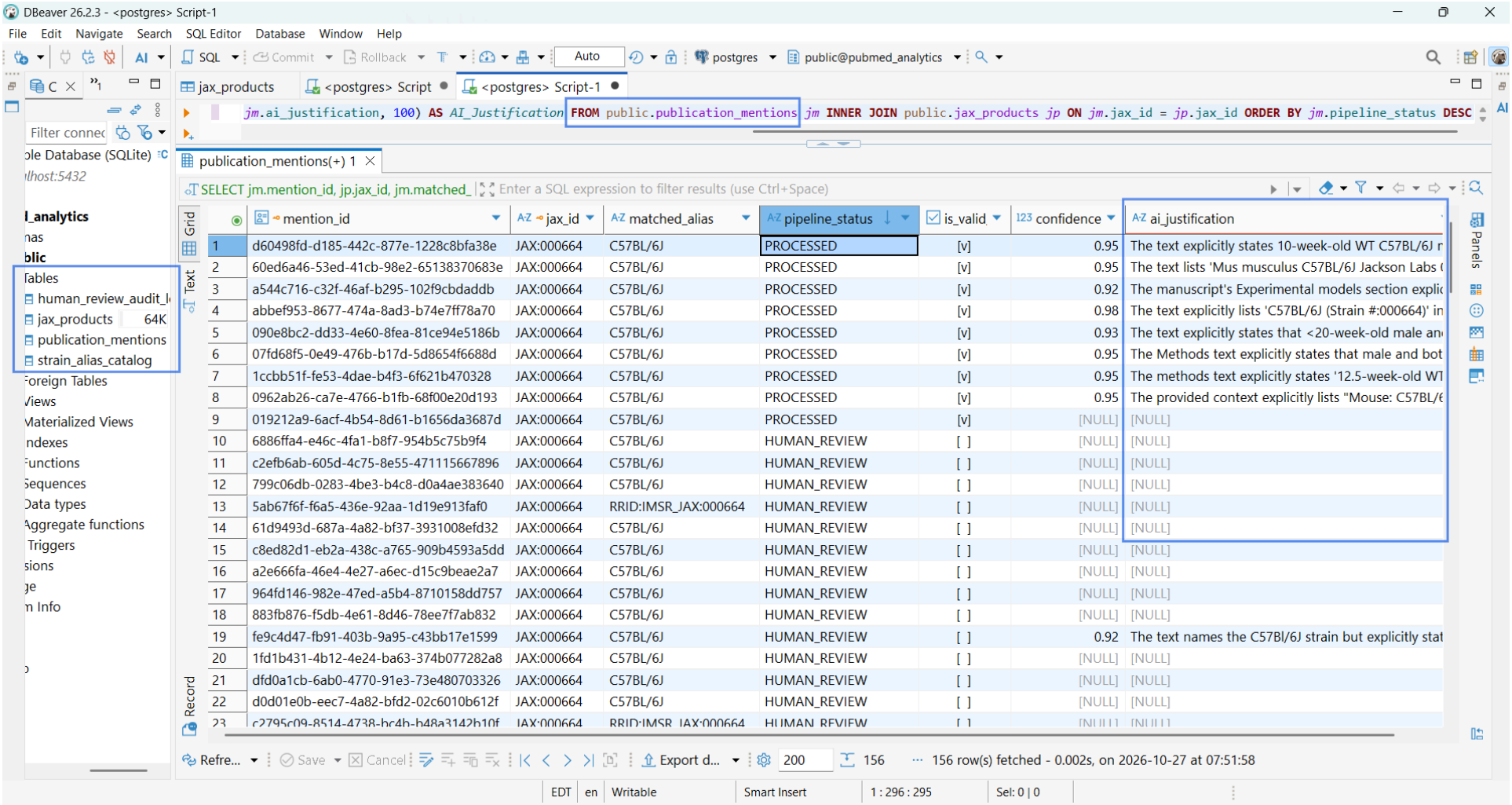


Figure 13: DBeaver interface showing query results from the “pubmed_analytics” database. These are examples of actual agentic evaluations generated by OpenAI GPT and captured into the PostgreSQL database.

```
SELECT distinct jm.ai_justification AS AI_Justification FROM public.publication_mentions jm
```

ai_justification
The text lists 'Mus musculus C57BL/6J Jackson Labs 010724' under Experimental models/Organisms/strains, indicating current use of the C57BL/6J strain (with a Jackson Labs catalog reference) in the study; a TRIM21 -/- C57BL/6J line from Jackson Labs is also listed.
The provided context explicitly lists "Mouse: C57BL/6J mice Jackson Laboratory RRID:IMSR_JAX:000664" under Experimental models for this study, indicating current use of the JAX strain JAX:000664. The normalized term maps uniquely to a single JAX candidate, so the mention is unambiguous and active.
The text names the C57Bl/6J strain but explicitly states “Wild-type C57Bl/6J mice were purchased from Charles River Laboratories,” so the animals were obtained from Charles River, not The Jackson Laboratory. The authors explicitly annotate JAX-sourced lines elsewhere (e.g., “Gata1 tm6Sho/J (JAX, 005653)”), indicating they mark JAX provenance when applicable.
The methods text explicitly states '12.5-week-old WT C57BL/6J ... mice (6 in each group) were inoculated...' indicating current experimental use of the C57BL/6J (JAX:000664) strain in this study.
The Methods text explicitly states that male and both male and female C57BL/6J mice (8–15 weeks old) were used in the reported experiments, indicating current use of this strain. The normalized term maps uniquely to JAX:000664 (unambiguous).
The text refers to C57BL/6JCrl mice purchased from Charles River and C57BL/6J background animals provided by a professor; there is no indication the animals were sourced from The Jackson Laboratory (JAX:000664). Therefore the matched string is not a JAX product mention.
The text explicitly states 10-week-old WT C57BL/6J mice were used in a passive transfer experiment (age, group size, treatments, infection and euthanasia described). The normalized term maps uniquely to JAX:000664 (MGI:3028467), so this is an active, current use.
The manuscript's Experimental models section explicitly lists 'Mus musculus C57BL/6J Jackson Labs 010724' and 'Mus musculus C57BL/6J TRIM21 -/- Jackson Labs 010724', indicating current use/acquisition of the Jackson Laboratory C57BL/6J strain (JAX:000664) in the study.
The text explicitly lists 'C57BL/6J (Strain #:000664)' in the Animal studies section and describes active use (mice were housed, genotyped, and injected), indicating current experimental use of the JAX strain JAX:000664. The normalized term maps uniquely to this JAX/MGI identifier, so the mention is a valid, active JAX product use.
The text explicitly states that <20-week-old male and female C57BL/6J mice were used and divided into experimental groups for the HFpEF study, indicating current experimental use. Metadata shows an unambiguous mapping to JAX:000664 (MGI:3028467), supporting a valid JAX strain mention.

Figure 14: DBeaver interface showing query results from the “human_review_audit_log” table.

